pharmaniaga®

Pethidine Hydrochloride

50mg/mL Injection & 100mg/2mL Injection

COMPOSITION
Each ampoule contains Pethidine Hydrochloride 50mg/mL.

PRODUCT DESCRIPTION
A colourless to pale yellow solution.

PHARMACODYNAMICS
Pethidine is a narcotic analgesic that can be used for the relief of most types of moderate to severe pain. It interacts with stereospecific opiate receptors in the CNS and other tissues. Analgesia is produced primarily through an alteration in emotional response to pain. The relief of pain is fairly specific; other sensory modalities are essentially unaffected and mental processes are not impaired (unlike anaesthetics) except when given in large doses or to unusually susceptible individuals.

PHARMACOKINETICS
Absorbed by all routes of administration. After an IV dose of 0.8 mg/kg to normal subjects, plasma concentrations of about 0.2 µg/mL are obtained in 1 hour falling to 0.1 µg/mL after a further 2 hours; after an IM dose of 100 mg, plasma concentrations of about 0.5 µg/mL are attained in about 1 hour; the administration of anaesthetics appears to cause up to a two-fold increase in plasma concentration of unchanged drug; similarly plasma concentration appear to vary according to age, concentration being higher in older subjects.
Rapidly and extensively distributed throughout the tissues; volume of distribution, 200-300 liters; it readily crosses the placenta, is secreted in milk, and taken up to some extend by red blood cells; 40 to 65% bound to plasma proteins.

Metabolic reactions: N-Demethylation, hydrolysis, Glucuronic Acid conjugation and N-oxidation. Excretion: Influenced by pH value by of urine. In acid urine, 10 to 30% of dose is excreted unchanged, 10 to 35% is excreted as Norpethidine, and about 1% is excreted as Pethidine N-oxide; in alkaline urine less than 5% is excreted as Pethidine and Norpethidine. In normal subjects, about 70% of dose is excreted in the urine in 24 hours, up to 10% as unchanged drug, 10% as Norpethidine, 20% as Pethidinic Acid, 16% as conjugated Pethidinic Acid, 8% as Norpethidinic Acid, and 10% as conjugated Norpethidinic Acid.

INDICATION
Moderate to severe pain, obstetric analgesia, pre-operative analgesia, enhancement of anaesthesia, for basal narcosis with phenothiazines.

RECOMMENDED DOSAGE
By SC or IM injection: 25 - 100 mg repeated after 4 hours.
Child: IM: 0.5 - 1.5 mg/kg.
By slow IV injection: 25 - 50 mg, repeated after 4 hours.
Obstetric analgesia: By SC or IM injection, 50 - 100 mg, repeated 1 - 3 hours later if necessary; max. 400 mg in 24 hours.
Elderly and debilitated patients: Initial doses should not exceed 25 mg by injection as elderly and debilitated patients are likely to be particularly sensitive to the central depressant effects of the drug.
Children: Single dose of 0.5 - 2 mg/kg body weight intramuscularly. This dose may be repeated if clinically necessary but it should not be repeated more often than four hourly.

ROUTE OF ADMINISTRATION
By SC, IM or Slow IV Injection

CONTRAINDICATIONS
Severe liver disease, acute alcoholism, bronchial asthma, increased intracranial pressure, respiratory depression, patients who have received MAO inhibitors within 14 days, hypotension and during pregnancy and breast feeding. Pethidine is also contraindicated in patients with a history of hypersensitivity or idiosyncratic response to this drug.

WARNINGS AND PRECAUTIONS
Use with caution in patients with liver disease, decreased respiratory reserve or those receiving other CNS depressant drugs. Should not be used in the presence of Phenobarbitone or any other enzyme inducing agents. May produce dependence. Administration during labour may cause respiratory depression in the newborn infant. Use with caution in patients taking MAO inhibitors. Must be given cautiously to patients with supraventricular tachycardia. Not usually given pre-operatively to children under 1 year of age. Should be given with extreme care to newborn or premature infants, in elderly and debilitated patients.

Pethidine should be used with caution in patients with head injuries, severe hepatic or renal impairment, biliary tract disorders, hypothyroidism, adrenocortical insufficiency, shock, prostatic hypertrophy. Caution is also required in patients exhibiting acute alcoholism, raised intracranial pressure or convulsive disorders. Pethidine depresses respiratory function and should be administered with particular care in patients with respiratory insufficiency. Concomitant administration of pethidine with Phenothiazines may induce severe hypotension. Pethidine induced neurotoxicity may be seen in patients with renal failure, cancer or sickle anaemia, during concomitant administration with anticholinergics or during chronic administration of increasing pethidine doses. Patient should be instructed to avoid alcohol while under treatment, since the individual response cannot be foreseen. Pethidine may modify patient's reaction (driving ability, operation of machinery, etc) to a varying extent, depending on dosage, administration and individual susceptibility. Repeated administration of pethidine may induce tolerance to the drug with a tendency to increasing dosage requirements to obtain the desired effect, or to physical and psychological dependence of the morphine type, with the development of withdrawal symptoms after abrupt cessation of therapy. Cross-tolerance between narcotic analgesics can occur.

Risks from Concomitant Use with Benzodiazepines
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Pethidine with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Pethidine is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs
Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of pethidine hydrochloride with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue pethidine hydrochloride if serotonin syndrome is suspected.

Adrenal Insufficiency
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal

insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Post-marketing Experience).

INTERACTIONS WITH OTHER MEDICAMENTS
- Mostly all opioids (Analgesics) have interactions with alcohol and CNS depressants.
- Concurrent use between alcohol and pethidine may result in increased Central Nervous System depression, respiratory depression and hypotensive effect.

Benzodiazepines
Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Serotonergic Drugs
The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjument. Discontinue pethidine hydrochloride if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRls), serotonin and norepinephrine reuptake inhibitors (SNRls), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Incompatibilities
Solutions of pethidine hydrochloride are acidic. Pethidine is incompatible with barbiturate salts and with other drugs including aminophylline, amobarbital, floxacillin, furosemide, amylobarbitone sodium, ephedrine sulphate, heparin sodium, hydrocortisone sodium succinate (in a ratio of 2 parts of hydrocortisone sodium succinate to 1 part of pethidine hydrochloride), methicillin sodium, methylprednisolone sodium succinate, morphine sulphate, nitrofurantoin sodium, pentobarbitone sodium, phenobarbitone sodium, phenytoin sodium, oxytetracycline hydrochloride, thiamylal sodium, thiopental sodium, sodium bicarbonate, sulphadiazine sodium, sodium iodide, sulphafurazole diethanolamine or thiopentone sodium. Incompatibility has also been observed between pethidine hydrochloride and acyclovir sodium, imipenem, frusemide, idarubicin, allopurinol, amphotericin B cholesteryl sulphate complex and cefepime.

Colour changes or precipitation have been observed on mixing pethidine with the following drugs, minocycline hydrochloride, tetracycline hydrochloride, cefoperazone sodium, mezlocillin sodium, nafcillin sodium and liposomal doxorubicin hydrochloride.

Pethidine hydrochloride also has been reported to be incompatible with solutions containing potassium iodide, aminosalicyclic acid, and salicylamide.

PREGNANCY AND LACTATION
Pregnancy
There is inadequate evidence of safety in human pregnancy, but the drug has been in widely used for many years without apparent ill consequence. As with all drugs during pregnancy care should be taken in assessing the risk to benefit ratio.

Lactation
No adverse effects have been seen in breast-feeding infants whose mothers were given pethidine.

SIDE EFFECTS
Constipation (by reduction of intestinal motility), respiratory depression, cough suppression, urinary retention, nausea, and tolerance and are liable to cause dependence.
Other side effects: Mild euphoria, breath shortness, face edema, dizziness, nausea and vomiting, urticaria and other skin rashes, hypotension and also central nervous system excitation.

Post marketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

SYMPTOMS AND TREATMENT OF OVERDOSE
Symptoms: Dizziness, nausea, vomiting, syncope, euphoria, hallucinations, headache, vasolidation and hypotension. Muscular twitching, tremors, hyperactive reflexes, dilated pupils and convulsions followed by respiratory depression and coma; local reactions.
Treatment: If the drug has been ingested, empty stomach by aspiration and lavage. Intensive supportive therapy may be necessary to correct respiratory failure and shock. Use specific antagonist, naloxone hydrochloride - 400 μg IV repeated at intervals of 2 - 3 minutes if necessary. In children, a dose of 5 - 10 μg/kg body weight may be given. Nalorphine hydrobromide and levallorphan tartrate have also been used. The circulation should be maintained with infusions of Dextrose Injection, and suitable electrolyte solutions. Small doses of suxamethonium have also been suggested to control convulsions.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE
Pethidine may cause drowsiness, impair the mental and/or physical abilities required for driving or for operating machinery. If affected patients should be advised accordingly and warned not to drive or operate machinery.

This medicine can impair cognitive function and can affect a patient's ability to drive safely. This class of medicine is in the list of drugs included in regulations under 5a of the Road Traffic Act 1988. When prescribing this medicine, patients should be told:
- The medicine is likely to affect your ability to drive
- Do not drive until you know how the medicine affects you
- It is an offence to drive while under the influence of this medicine
- However, you would not be committing an offence (called 'statutory defence') if:
  - The medicine has been prescribed to treat a medical or dental problem and
  - You have taken it according to the instructions given by the prescriber and in the information provided with the medicine and
  - It was not affecting your ability to drive safely

STORAGE CONDITIONS
Store below 30°C. Protect from light. Retain in carton until time of use.
Do not use if solution contains growth, particles, turbidity or if there is any change of appearance.

SHELF LIFE
3 years from date of manufacture.

DOSAGE FORMS AND PACKAGING AVAILABLE:
Pethidine Hydrochloride 50mg/mL Injection
10 x 1 mL ampoule (clear glass)
Pethidine Hydrochloride 100mg/2mL Injection
10 x 2 mL ampoule (clear glass)

PRODUCT REGISTRATION HOLDER /MANUFACTURER
Pharmaniaga Lifescience Sdn Bhd (198201002939)
Lot 7, Jalan PPU 3,
Taman Perindustrian Puchong Utama,
47100 Puchong, Selangor Darul Ehsan, Malaysia

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